

MATH0046 Calculus in Several Dimensions

<i>Year:</i>	2026–2027
<i>Code:</i>	MATH0046
<i>Level:</i>	4 (UG)
<i>Normal student group(s):</i>	UG: Y1 Mathematics programmes
<i>Value:</i>	15 credits (7.5 ECTS credits)
<i>Term:</i>	2
<i>Assessment:</i>	85% examination, 15% coursework
<i>Normal Pre-requisites:</i>	MATH0045 (Calculus and Linear Algebra)
<i>Lecturer:</i>	Dr J Hofmann

Course Description and Objectives

This course is a continuation of the MATH0045 course. The emphasis is on developing technique rather than full rigour. Topics covered include optimisation with a constraint, complex numbers, differential equations, multiple integrals, eigenvalues and eigenvectors and quadratic forms.

Recommended Texts

- G Stephenson, *Mathematical Methods for Science Students* (Longman)
- Riley, Hobson and Bence, *Mathematical Methods for Physics and Engineering* (Cambridge)

Detailed Syllabus

- Review of unconstrained optimisation: Optimisation problems with a constraint. Use of Lagrange multipliers. Examples of optimisation problems with inequality constraints.
- Complex numbers: The Argand diagram, modulus and argument, complex-conjugate. Cartesian and polar forms. De Moivre's Theorem. Roots of polynomials. Introduction to the Fundamental Theorem of Algebra.
- Review of the linear independence of vectors and determinants of larger matrices: Eigenvalues and eigenvectors of a matrix. Diagonalisation of a matrix. Properties of symmetric matrices. Quadratic forms.
- First order differential equations: Separation of variables. General linear equation and integrating factor. First order linear systems with constant coefficients.
- Multiple integrals: Reduction to a repeated integral. Change of the order of integration in a repeated integral. Cylindrical and spherical polar co-ordinates. Change of variables and the Jacobian.
- Special functions: Gamma and Beta functions, Stirling's formula.